

CHRONIX EDU

Master ToDo List

One Task · One Prompt · One Review

Version 1.0 | Chronix Technology Limited | Target: September 7, 2025

Total tasks	47 tasks across Foundation, Phase 1, and Phase 2
Stretch tasks	9 additional tasks if Phase 3 timeline is beaten
Prompt reference	Each task maps to exactly one prompt in the Build Prompt Guide
Test reference	Each task maps to exactly one test in The Test document
Rule	Complete one task. Run its test. Pass the test. Move to next task.
Blocked rule	A BLOCKED task stops all work. Do not skip it. Fix the blocker first.
Status values	TODO · IN PROGRESS · DONE · BLOCKED

FOUNDATION PHASE

Database · Auth · Multi-tenancy · Project Setup

Weeks 1–3 · 8 Tasks · Prompts P1–P8

Nothing else can be built until every task in this section is complete and tested. These tasks have no shortcuts.

#	Task	Prompt	Test	Phase	Status
F1	Scaffold monorepo — /apps/web, /apps/api, /packages/shared	P1	T1	Foundation	TODO
NOTE	Install npm workspaces. Configure path aliases (@/) in tsconfig for both apps. Set up ESLint, Prettier, and Husky. Create .env.example with all required variables. Verify both apps start without errors before moving to F2.				
F2	Write full PostgreSQL schema — all 16 core tables + title field on users	P2	T2	Foundation	TODO
NOTE	Add title VARCHAR(20) NULLABLE to the users table (Dr., Prof., Engr., Mrs., Mr., Chief, etc.). This was added during the dashboard review session. Save as /apps/api/src/migrations/001_initial_schema.sql.				
F3	Enable Row Level Security on all tables — write and test isolation policies	P3	T3	Foundation	TODO
NOTE	CRITICAL: Run the cross-tenant isolation test before marking DONE. School A must return zero results when queried with School B's JWT. No exceptions.				
F4	Build authentication — JWT with custom claims (school_id, role, title, user_id)	P4	T4	Foundation	TODO
NOTE	JWT claims must include title field alongside role and school_id. verifyToken and requireRole middleware saved to /apps/api/src/middleware/auth.ts.				
F5	Build school onboarding and settings API	P5	T5	Foundation	TODO
NOTE	Auto-seed default grading scale and assessment components on school creation. Default grading: A=70-100, B=60-69, C=50-59, D=40-49, F=0-39. Default components: CA1 10%, CA2 10%, Mid-Term 10%, Exam 70%.				
F6	Build academic session and term management API — /current-context with 60s cache	P6	T6	Foundation	TODO
NOTE	Add partial unique index: CREATE UNIQUE INDEX one_current_session ON academic_sessions(school_id) WHERE is_current = true. First session seeded for Child Prime Onyx: First Term Sep–Dec 2025.				
F7	Build class, subject, and teacher assignment APIs — lock mode per teacher account	P7	T7	Foundation	TODO
NOTE	Add teacher_mode ENUM('class','subject') to users table. Set by admin at user creation. Class teacher = class mode (primary). Subject teacher = subject mode (secondary). System uses this to determine score entry UI layout.				
F8	Build assessment configuration engine — 4-level resolution, weight validation, 423 lock	P8	T8	Foundation	TODO

NOTE

Test all 4 resolution levels: subject+class_level, subject-only, class_level-only, school-wide default. Seed Child Prime Onyx with default config on onboarding.

PHASE 1 — MVP

Students · Scores · Results · Report Cards · Dashboards

Weeks 4–10 · 13 Tasks · Prompts P9–P20b

Phase 1 ends with a product that can be demonstrated to Child Prime Onyx School. A principal must be able to approve and publish results. A teacher must be able to enter scores. Report cards must generate with school branding.

#	Task	Prompt	Test	Phase	Status
M1	Build student registration API — auto admission number, parent accounts, promote endpoint	P9	T9	Phase 1	TODO
NOTE	Admission number format: SCH-[YEAR]-[4-digit-sequence]. Parent accounts linked via parent_students junction. promote endpoint creates new student_classes row — never deletes old rows.				
M2	Build score entry API — validation, bulk entry, audit log with old value capture	P10	T10	Phase 1	TODO
NOTE	Three validations per score: (1) teacher is assigned to subject+class+term, (2) score does not exceed component max, (3) result_status is not approved or published. Capture old score before upsert for audit log.				
M3	Build result computation engine — weighted totals, grades, tied positions, at-risk	P11	T11	Phase 1	TODO
NOTE	Pure TypeScript — no database writes. Tied positions: two students at equal average both receive same position, next position is skipped. Export computeStudentSubjectResult(), computeClassResults(), getStudentsAtRisk().				
M4	Build result approval workflow — DRAFT > SUBMITTED > APPROVED > PUBLISHED	P12	T12	Phase 1	TODO
NOTE	validateStatusTransition() guard used on every endpoint. Submit checks all components are scored before allowing. Approve checks all subjects are submitted. Publish queues parent notification — does not send inline.				
M5	Build report card PDF generator — Puppeteer, Handlebars, Supabase Storage upload	P13	T13	Phase 1	TODO
NOTE	Template must include: school logo, name, motto, student photo, results table, teacher comment, principal remark, attendance placeholder, resumption date, school stamp, promotion status. Test with 15+ subjects and long names.				
M6	Build principal dashboard API — 5-minute cache on all endpoints	P14	T14	Phase 1	TODO
NOTE	Principal is addressed by title if available: Dr., Prof., Engr., etc. Greeting: 'Good morning, Dr. Okafor' not 'Good morning, Okafor'. Cache each endpoint 5 minutes — never compute positions live per request.				
M7	Build teacher dashboard API — scoped to teacher's own assignments only	P15	T15	Phase 1	TODO
NOTE	Teacher is addressed by title if available. Dashboard groups pending work by subject-across-classes for subject-mode teachers, and by class for class-mode teachers.				

M8	Build admin settings panel UI — identity, academic structure, grading scale pages	P16	T16	Phase 1	TODO
NOTE	<i>Identity page: logo upload, colour picker with live portal header preview. Academic structure: session/term management with activation confirmation modal. Grading scale: contiguous band validation, warn if published results exist.</i>				
M9	Implement settings locking and audit trail — 423, 400, 409 guards + logSettingsChange()	P16b	T16b	Phase 1	TODO
NOTE	<i>Every PATCH to school_settings writes to audit_logs via logSettingsChange(). Padlock icons on locked fields in UI. Checkbox gate before saving grading scale when published results exist.</i>				
M10	Build assessment configuration UI — dynamic component builder, weight validator, resolution preview	P17	T17	Phase 1	TODO
NOTE	<i>Weight total computed client-side in real time. Green at 100%, red at any other value. Save button disabled until total equals 100. Locked configs show padlock icon and disabled state — teacher cannot edit after scores entered.</i>				
M11	Build user management UI — role-specific create modal, teacher mode set at creation	P18	T18	Phase 1	TODO
NOTE	<i>Teacher creation modal includes teacher_mode toggle: Class teacher (primary) or Subject teacher (secondary). This sets teacher_mode on the users record. Cannot be changed after creation without admin intervention.</i>				
M12	Build score entry UI — subject-first mode for secondary, class-first for primary	P19	T19	Phase 1	TODO
NOTE	<i>UI mode determined by teacher_mode on user account. Subject-mode: assignment cards at top, class tabs below, submit button says Submit [Class] for approval. Class-mode: class selector first, subject selector second. No toggle shown to teacher.</i>				
M13	Build principal result approval and report card UI — dropdown class selector, two-step publish	P20	T20	Phase 1	TODO
NOTE	<i>Class selector is a dropdown — not a horizontal scroll grid. Approve button active only when all subjects are Submitted. Publish button requires two-step confirmation. Principal addressed by title in all greetings.</i>				

PHASE 2 — OPERATIONS

Attendance · Parent Portal · Messaging · Offline PWA

Weeks 11–13 · 13 Tasks · Prompts P21–P31

Phase 2 is the launch target for September 7th. If any task in this phase cannot be completed before the deadline, cut in this order: assignments first, behaviour second, notification settings third. Attendance and parent portal are non-negotiable.

#	Task	Prompt	Test	Phase	Status
O1	Build attendance schema and API — upsert, 3-day absent alert, monthly summary	P21	T21	Phase 2	TODO
NOTE	<i>Attendance is always visible to parents regardless of result publication status. Alert fires when a student has 3+ absents in 7 days — queues parent notification immediately, not batched.</i>				
O2	Build attendance UI — toggle buttons for teachers, calendar view, principal overview	P22	T22	Phase 2	TODO
NOTE	<i>Toggle buttons (P/A/L/E), not dropdowns. Teachers can correct same-day attendance. 30-day calendar view per student. Principal view shows school-wide attendance % per class and chronic absenteeism list.</i>				
O3	Build parent portal — mobile-responsive, multi-child selector, all 5 sections	P23	T23	Phase 2	TODO
NOTE	<i>Child selector hidden if parent has only one child. Full component breakdown visible: CA1, CA2, Mid-Term, Exam shown individually per subject. Attendance always visible. Fee card always visible with Paystack button disabled ('Coming soon') until Phase 3. Bottom nav: Home, Results, Attendance, Messages, Fees.</i>				
O4	Build student portal — results with term selector, attendance, notices	P24	T24	Phase 2	TODO
NOTE	<i>Student sees only their own data. Results only visible after principal publishes. Assignments tab visible as placeholder even if P25 not yet built.</i>				
O5	Build assignment module — create, deadline enforcement, grade and feedback	P25	T25	Phase 2	TODO
NOTE	<i>POST submission returns 400 after due_date — no exceptions. Teacher UI shows all submissions per assignment with grading interface. Student sees grade and feedback only after teacher has graded.</i>				
O6	Build behaviour and discipline module — auto parent notification, suspension = immediate	P26	T26	Phase 2	TODO
NOTE	<i>Suspension-severity incidents bypass the notification queue and send immediately. Behaviour card in parent portal shows incident count this term with detail view. Clean record shows: No incidents recorded this term.</i>				
O7	Build messaging module — structured inbox, announcements, notification worker	P27	T27	Phase 2	TODO
NOTE	<i>Role-pair enforcement: parent can message their child's teacher or principal only. Worker runs every 30 seconds. Email via SendGrid only at launch — Termii SMS wired in Phase 3. Notification bell in header with unread badge count.</i>				

O8	Build offline PWA — Dexie.js queue for attendance and score entry, auto-sync	P28	T28	Phase 2	TODO
NOTE	Test offline: Chrome DevTools > Network > Offline. Mark attendance offline. Reconnect. Verify sync. Three-state indicator: green (online synced), yellow (syncing), red (offline, N items pending). Must pass before Phase 2 sign-off.				
O9	Build notification settings UI — event toggles, alert threshold, email test button	P29	T29	Phase 2	TODO
NOTE	SMS sender name field present but greyed out at launch — activates in Phase 3. Email test button sends test to admin email to verify SendGrid. Attendance alert threshold defaults to 3 absences in 7 days.				
O10	Build registrar module — multi-step admissions, promotion manager, transcript PDF	P30	T30	Phase 2	TODO
NOTE	Promotion manager locked to registrar and super_admin roles only. Promote endpoint creates new student_classes row — never deletes historical rows. Transcript PDF shows all terms across all sessions.				
O11	Run Phase 2 integration tests — 5 scenarios must all pass	P31	T31	Phase 2	TODO
NOTE	5 tests: offline sync, parent data isolation, notification pipeline with SendGrid mock, behaviour auto-alert, assignment deadline enforcement. All 5 must pass before any Phase 3 work begins.				
O12	Seed Child Prime Onyx School data — students, classes, teachers, terms	P1	T32	Phase 2	TODO
NOTE	This is a non-coding task. Collect from Child Prime Onyx School: full student list with names and DOB, class names, teacher names and subjects, staff titles (Dr., Mrs., Mr., etc.). Import to database 2 weeks before September 7th.				
O13	Production deployment — Vercel (web) + Railway (API) + custom domain	P40	T33	Phase 2	TODO
NOTE	Domain: edu.chronixtechnology.com. Configure Vercel for Next.js frontend. Configure Railway for Express API. Set all environment variables in both platforms. Run startup env var validation. Verify SSL certificate is active.				

PHASE 3 — STRETCH

Fees · Paystack · Analytics · Timetable · SMS

Post September 7th · If timeline beaten · 9 Tasks · Prompts P32–P40

Phase 3 tasks begin only after all Phase 2 tasks are DONE and the system is live at Child Prime Onyx School. Do not start Phase 3 during the Phase 2 deadline sprint.

#	Task	Prompt	Test	Phase	Status
S1	Build fees and billing API — fee structures, invoice generation, manual payment	P32	T34	Phase 3	TODO
NOTE	Fee card and Paystack button already exist in parent portal UI from Phase 2. This task wires the data. Invoice generation creates fee_invoices rows for all students in the term.				
S2	Integrate Paystack — initiate, callback, webhook, activate Pay Now button	P33	T35	Phase 3	TODO
NOTE	Activate the Pay Now button in parent portal (remove 'Coming soon' state). Always verify transactions server-side — never trust callback URL parameters. Webhook is the primary confirmation mechanism.				
S3	Build bursar dashboard — fee management, PDF receipt, collection reports	P34	T36	Phase 3	TODO
S4	Build advanced analytics — snapshots, nightly cron, 5 chart types	P35	T37	Phase 3	TODO
NOTE	Analytics snapshots generated nightly at 2am. Never query raw scores live for dashboard. Principal analytics page added to sidebar navigation.				
S5	Build timetable module — slot management, clash detection, drag-and-drop builder	P36	T38	Phase 3	TODO
NOTE	Timetable tab appears in student and teacher sidebars once this task is DONE. 409 on teacher or class clash with clear description of what is clashing.				
S6	Integrate Termii SMS — activate SMS sender in notification settings	P37	T39	Phase 3	TODO
NOTE	Activate SMS sender name field in notification settings UI (was greyed out). Per-school Termii API key stored in school_settings — not a global env variable. 3 SMS per user per day throttle.				
S7	Add automated fee reminders — weekly Monday 8am cron + manual bursar trigger	P38	T40	Phase 3	TODO
S8	Build report card settings UI — template selector, signature upload, live preview	P39	T41	Phase 3	TODO
NOTE	Live preview uses dummy student data with real school branding. Never use a real student's data for preview.				
S9	Phase 3 readiness audit — env vars, query performance, rate limiting, logging	P40	T42	Phase 3	TODO
NOTE	EXPLAIN ANALYZE on 5 critical queries. Artillery load test: all endpoints under 2000ms at 50 concurrent users. Winston logging active. Rate limiting: 100/min general, 5/min auth.				

Schema Changes Confirmed During Pre-Build Session

These changes were confirmed during the dashboard prototype review and must be applied to the schema in task F2. They are not in the original architecture document.

Table	Change	Reason
users	Add title VARCHAR(20) NULLABLE	<i>Principals, teachers addressed by Dr., Prof., Engr., Mrs., Mr., Chief, etc. when available</i>
users	Add teacher_mode ENUM('class','subject') NOT NULL DEFAULT 'subject'	<i>Determines score entry UI layout. Set by admin at user creation. Class = primary teacher, Subject = secondary teacher.</i>
school_settings	Seed default grading scale on school creation (JSONB)	<i>A=70-100, B=60-69, C=50-59, D=40-49, F=0-39. Promotion cutoff = 40. Configurable per school.</i>
school_settings	Seed default assessment components on school creation (JSONB)	<i>CA1 10%, CA2 10%, Mid-Term 10%, Exam 70%. Configurable per school.</i>
school_settings	Seed default academic calendar on school creation (JSONB)	<i>3 terms: First Term Sep-Dec, Second Term Jan-Apr, Third Term May-Jul.</i>

UI Decisions Confirmed During Pre-Build Session

Decision	What to build
User greeting format	Use title if available: 'Good morning, Dr. Okafor'. Never first name only. Fall back to full name if no title stored.
Principal result grid	Dropdown class selector — not horizontal scroll. Class overview stays in stat cards. Grid shows one class at a time.
Teacher score entry mode	Mode locked per account (teacher_mode field). Set by admin at user creation. No toggle shown to teacher in the UI.
Score submission granularity	Per class per subject. Never bulk-submit all classes at once. Submit button: 'Submit JSS 2A for approval'.
Parent child selector	Hidden if parent has only one child. Shown as dropdown chevron if two or more children.
Parent component breakdown	Full breakdown visible: CA1, CA2, Mid-Term, Exam shown individually. Helps parents identify where child needs support.
Parent fee card	Always visible from launch. Paystack Pay Now button present but disabled with 'Coming soon' label. Activates in Phase 3.
Parent bottom navigation	5 tabs: Home, Results, Attendance, Messages, Fees. Fees tab always visible — activates in Phase 3.
Attendance visibility	Always visible to parents regardless of result publication status.
Teacher assignment overview	Assignment cards above the score grid on the same page. Not a separate page. Max 5 subjects per teacher.